

ChainProof: Decentralized Placement Tracking & Credential Verification System

1. Title of the Project

ChainProof: A Blockchain-Based Decentralized Placement Tracking and Credential Verification System

2. Introduction

Indian university placement processes suffer from a fundamental lack of transparency and accountability. Colleges self-report inflated placement statistics to regulatory bodies such as NAAC, students have no verifiable proof of their skills and achievements outside college control, and companies have no reliable way to validate candidate credentials without contacting institutions directly.

The result is a three-way blame cycle: colleges blame students for poor participation, students blame colleges for not bringing companies, and companies spend significant resources screening unverified candidates. No neutral record of any party's actual effort exists.

DigiLocker, India's existing digital document infrastructure, partially addresses certificate storage but remains centralized, government-controlled, and limited to official documents. It cannot capture granular achievements such as workshop attendance, internship completions, project contributions, or recruitment outcomes. Critically, it cannot prevent institutions from manipulating the data they submit.

ChainProof addresses this gap by building a decentralized application (DApp) on the Ethereum blockchain where every placement-related action — credential issuance, company visits, interview outcomes, offer letters — is recorded immutably by the actual party performing it. No single authority controls the data. Placement statistics are computed automatically from on-chain records, making manipulation structurally impossible.

3. Objectives

1. To design and deploy a role-based smart contract system that registers students, colleges, and companies as verified actors on the Ethereum blockchain.
2. To implement a tamper-proof credential issuance mechanism where colleges and companies issue verifiable credentials directly to student wallet addresses.
3. To build an automated placement statistics engine that calculates real placement percentages from on-chain data, eliminating self-reporting by institutions.
4. To develop a public accountability dashboard that displays the activity of all three actors transparently and in real time.
5. To deliver a complete, deployable DApp with a React-based frontend connected to smart contracts on the Ethereum Sepolia testnet.

4. Scope of the Project

In scope:

- Smart contract development in Solidity for actor registration, credential issuance, and placement tracking
- Deployment on Ethereum Sepolia testnet
- React-based frontend with MetaMask wallet integration
- Role-specific dashboards for students, colleges, and companies
- Public placement accountability dashboard
- IPFS integration for off-chain document storage via Pinata

Out of scope (Phase 2):

- Proof of Merit Token (PMT) incentive and reward system
- Stake-based dispute resolution mechanism
- Reputation scoring algorithm
- NAAC direct data feed integration
- Mobile application
- Mainnet deployment

5. Literature Review

#	Authors	Title	Journal/Conference	Year
1	Nakamoto, S.	Bitcoin: A Peer-to-Peer Electronic Cash System	bitcoin.org	2009
2	Wood, G.	Ethereum: A Secure Decentralised Generalised Transaction Ledger	Ethereum Yellow Paper	2014
3	Turkanović et al.	EduCTX: A Blockchain-Based Higher Education Credit Platform	IEEE Access	2018
4	Zheng et al.	Blockchain Challenges and Opportunities: A Survey	Int. Journal of Web and Grid Services	2018
5	Alammary et al.	Blockchain-Based Applications in Education: A Systematic Review	Applied Sciences	2019
6	Poorani et al.	DIGICERT: A Secured Digital Certificate Application using Blockchain through Smart Contracts	IEEE ICCES	2019
7	Mishra et al.	Implementation and Analysis of Blockchain Based DApp for Secure Sharing of Students' Credentials	IEEE CCNC	2020
8	Vidal et al.	Revocation Mechanisms for Academic Certificates Stored on a Blockchain	CISTI	2020
9	Nurhaeni et al.	Adoption of Upcoming Blockchain Revolution in Higher Education: Its Potential in Validating Certificates	IEEE ICIC	2020
10	Xie et al.	Ethereum-Blockchain-Based Technology of Decentralized Smart Contract Certificate System	IEEE Access	2020
11	Raimundo & Rosário	Blockchain System in the Higher Education	European Journal of Investigation in Health, Psychology and Education	2021

12	Rhemananda et al.	Blockchain Technology to Support Employee Recruitment and Selection in Industry 4.0	Lecture Notes in Networks and Systems	2021
13	Rupa et al.	Industry 5.0: Ethereum Blockchain Technology Based DApp Smart Contract	Mathematical Biosciences and Engineering	2021
14	Rani et al.	A Systematic Study on Blockchain Technology in Education: Initiatives, Products, Applications, Benefits, Challenges	Computing	2023
15	Kabashi et al.	Trustworthy Verification of Academic Credentials through Blockchain Technology	iJOE	2024
16	Cuya et al.	Blockchain in Higher Education: Advancing Security, Verification, and Trust in Academic Credentials	Nanotechnology Perceptions	2024
17	Xu, Y.	Development of Blockchain-Based Academic Credential Verification System	ResearchGate	2024
18	Berrios Moya et al.	A Zero-Knowledge Proof-Enabled Blockchain-Based Academic Record Verification System	Sensors, MDPI	2025
19	[]	Secure and Efficient Graduate Employment: A Consortium Blockchain Framework with IPFS for Privacy-Preserving Resume Management	PLOS One	2024
20	[]	Blockchain-Based Decentralized Micro-Accreditation for Lifelong Learning	Journal of Educational Technology	2024

6. Proposed Methodology

Phase 1 — Research & Design (Week 1) Study existing blockchain credential systems. Finalize smart contract architecture. Define data structures, actor roles, and contract interactions.

Phase 2 — Smart Contract Development (Week 2–3) Write and unit test all Solidity contracts using Hardhat. Deploy to local Hardhat network. Validate all functions and role-based access controls.

Phase 3 — Frontend Development (Week 4–5) Build React-based SPA using Vite. Implement MetaMask wallet connection. Build role-specific dashboards and public accountability view. Connect frontend to deployed contracts via ethers.js.

Phase 4 — Integration & Testnet Deployment (Week 6) Deploy contracts to Ethereum Sepolia testnet. Connect frontend to live contracts. Integrate IPFS via Pinata for document storage. End-to-end testing across all three actor roles.

Technologies and Tools:

- **Smart Contracts:** Solidity ^0.8.20
- **Development Framework:** Hardhat
- **Blockchain Network:** Ethereum (Sepolia Testnet)
- **Frontend:** React.js + Vite
- **Blockchain Interaction:** ethers.js
- **Wallet:** MetaMask
- **Decentralized Storage:** IPFS via Pinata
- **Testing:** Hardhat test suite (Mocha + Chai)
- **Version Control:** Git + GitHub

7. System Architecture

Three-layer architecture:

None

Layer 1 – Presentation (Frontend)

React SPA → MetaMask → Role detection → Dashboard routing

Student Dashboard | College Portal | Company Terminal | Public Dashboard

Layer 2 – Blockchain (Smart Contracts on Ethereum Sepolia)

ActorRegistry.sol → CredentialIssuer.sol → PlacementTracker.sol → StatsEngine.sol

Layer 3 – Storage (Decentralized)

IPFS via Pinata → document hashes stored on-chain → files stored off-chain

Actor interaction flow:

None

College registers → onboards students → brings companies

Company registers → posts visits → issues credentials to students

Student registers → receives credentials → builds verified profile

StatsEngine → reads all on-chain events → auto-calculates placement %

Public Dashboard → reads StatsEngine → displays tamper-proof statistics

8. Modules Description

Module 1: ActorRegistry.sol Handles registration of all three actor types — Student, College, Company — as wallet addresses with assigned roles. Prevents duplicate registration. Emits events on registration. Provides role-based access modifiers used by all other contracts.

Module 2: CredentialIssuer.sol Allows colleges and companies to issue verifiable credentials to registered student wallet addresses. Supports credential types: General, Shortlist, Interview, Offer, Rejection. Automatically marks students as placed when an Offer credential is issued. Stores IPFS hash of supporting documents on-chain.

Module 3: PlacementTracker.sol Records all placement process events — company visits announced by college, student applications, interview schedules, and final outcomes. Creates a permanent, timestamped, publicly visible log of all placement activity.

Module 4: StatsEngine.sol Reads data from ActorRegistry and CredentialIssuer to compute real-time placement statistics. Calculates placement percentage as placed students divided by total enrolled students — denominator cannot be manipulated by college.

Module 5: React Frontend Role-aware single-page application. Detects connected wallet role on load and routes to appropriate dashboard. Student dashboard shows credential timeline. College portal shows placement metrics and company visit log. Company terminal shows registered students and recruitment pipeline. Public dashboard shows institution-wise accountability data.

9. Feasibility Study

Technical Feasibility: All tools are open-source and freely available. Ethereum Sepolia testnet provides a free deployment environment. Hardhat, React, ethers.js, and Pinata are well-documented with large developer communities. The system is technically feasible within the defined scope.

Operational Feasibility: The system mirrors existing placement workflows — college contacts companies, students apply, companies interview. No new process is introduced. Actors perform the same actions they already do, now recorded on-chain.

Economic Feasibility: Development cost is zero — all tools are free and open-source. Testnet deployment costs no real ETH. Pinata offers a free IPFS tier sufficient for a prototype. Mainnet deployment in Phase 2 would require minimal ETH for gas fees.

10. Expected Outcome

A fully functional, deployed DApp on Ethereum Sepolia testnet where:

- Any wallet can register as a Student, College, or Company
- Colleges and companies can issue verifiable credentials to student wallets
- Placement statistics are computed automatically and publicly visible
- All three actor dashboards are operational and connected to live contracts
- A public accountability dashboard displays tamper-proof placement records

11. Tools and Technologies Used

Category	Tool
Smart Contract Language	Solidity ^0.8.20
Development Framework	Hardhat
Blockchain Network	Ethereum Sepolia Testnet
Frontend Framework	React.js + Vite
Blockchain Library	ethers.js
Wallet	MetaMask
Decentralized Storage	IPFS via Pinata
Testing Framework	Mocha + Chai (via Hardhat)
Version Control	Git + GitHub
IDE	Google Antigravity

12. Timeline

Week	Phase	Deliverable
Week 1	Research & Design	Architecture finalized, data structures defined
Week 2	Smart Contracts	ActorRegistry.sol + CredentialIssuer.sol written and tested
Week 3	Smart Contracts	PlacementTracker.sol + StatsEngine.sol written and tested
Week 4	Frontend	React SPA setup, MetaMask connection, routing
Week 5	Frontend	All four dashboards built and connected to contracts
Week 6	Deployment	Sepolia deployment, IPFS integration, end-to-end testing

13. Conclusion

ChainProof addresses a genuine and widespread problem in Indian university placement systems — opacity, manipulation, and zero accountability — by replacing self-reported data with immutable on-chain records generated by the actual parties involved. Unlike existing solutions such as DigiLocker, ChainProof is not controlled by any single authority. Placement statistics cannot be fabricated because they are computed directly from verified on-chain interactions. Students own their verified credential history. Companies build a transparent hiring record. Colleges are held accountable by math, not auditors. The system has direct relevance to NAAC accreditation processes and is extensible to a token-based incentive layer in Phase 2. ChainProof demonstrates that blockchain technology can solve real institutional accountability problems beyond financial applications.